Felipe López Montes

DATABASE ENGINEER | PostgreSQL · AWS

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PostgreSQL database engineer focused on performance optimization, access control and automation on Amazon RDS. Independently drives database improvement initiatives, from identifying problems to designing and implementing solutions, with experience on production instances up to 4 TB.

PROFESSIONAL EXPERIENCE

AIRCURY Remote

Database Technical Lead Oct 2025 – Present

- Strengthened **access controls** on 3 production RDS instances with individual accounts, IAM authentication and least privilege; led security auditing and remediation planning across 15 instances.
- Standardized PostgreSQL **maintenance deployment** across 4 production RDS instances by extending Lambda and Step Functions workflows and automating deployment with GitHub Actions.
- Built **ai-postgres** for growth tracking, forecasts, alerts, deterministic tuning recommendations and role/permission views; added an MCP server and Bedrock API for instance-aware natural-language questions.
- Led a **10-member Database Guild**, providing training and guidance on query optimization, indexing and database practices to teams working across 15 production RDS instances.

Data Engineer Mar 2024 – Oct 2025

- Reduced a client's monthly **AWS spend by ~30%** through PostgreSQL query and index tuning and data-pipeline optimization.
- Achieved **up to 60× faster queries**, prioritizing the top 10 by load in AWS Database Insights; used EXPLAIN and EXPLAIN ANALYZE to guide indexing and SQL rewrites.
- Cut daily **synchronization volume by ~98%** (~5M to ~100K records or fewer) using incremental reads, upserts and date filters instead of full refreshes, reducing CPU, memory use and runtime.
- **Reclaimed ~20%** of occupied PostgreSQL storage using pg_repack and historical-data purging; established weekly maintenance across 4 production RDS instances.
- Standardized 3 PostgreSQL schemas for UK Department for Education and Salesforce integrations by normalizing models and aligning fields with these source APIs.
- Delivered 3 PostgreSQL workshops for ~13 engineers on optimization, indexing, transactions and normalization.

Full-stack Web Developer Aug 2022 – Oct 2023

Developed React and PHP/Symfony features with PostgreSQL, integrating Salesforce and APIs using TDD/BDD.

STRATEGIC THINKING Remote

Full-stack Web Developer May 2021 – Dec 2021

Developed Vanilla JS features with Google Firebase Firestore.

TECHNICAL SKILLS

PostgreSQL: SQL · EXPLAIN ANALYZE · Indexing · Schema design · Data retention · pg_repack
AWS & security: RDS · Database Insights · IAM authentication · ECS · Bedrock
Automation & CI/CD: Lambda · Step Functions · GitHub Actions · Jenkins · Docker · Git
Development: PHP/Symfony · API integrations · Incremental synchronization · MCP

EDUCATION & LANGUAGES

University of Vigo 2016 – 2022
Bachelor's degrees in Computer Engineering and Business Administration.
Le Wagon, Zürich · Data Science Bootcamp Oct – Dec 2023

Languages: English C1 · French B2 · Spanish and Galician native